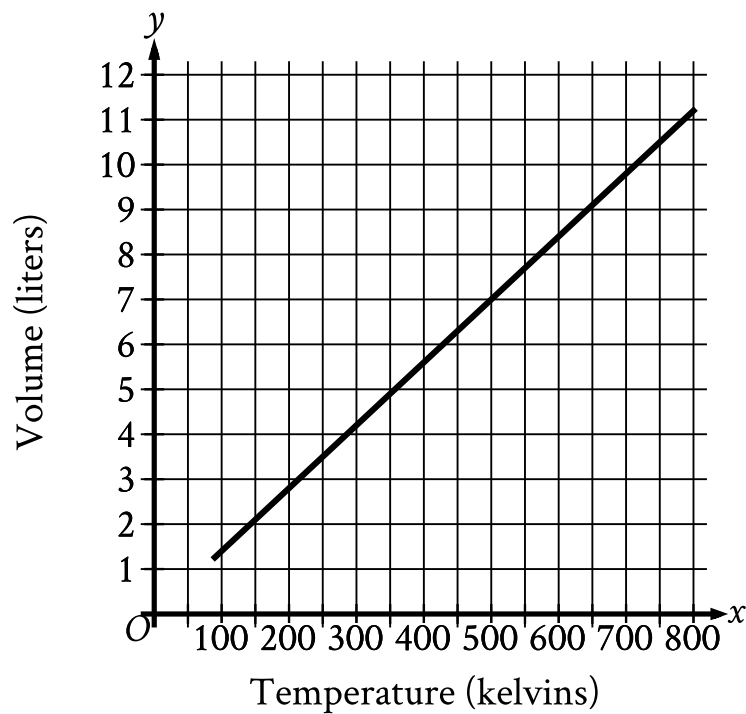


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

Hydrogen is placed inside a container and kept at a constant pressure. The graph shows the estimated volume y , in liters, of the hydrogen when its temperature is x kelvins.



What is the estimated volume, in liters, of the hydrogen when its temperature is **500** kelvins?

- Answer**
- A. 0
 - B. $\frac{7}{500}$
 - C. 7
 - D. $\frac{500}{7}$

Correct Answer: C

Rationale

Choice C is correct. For the graph shown, the x -axis represents temperature, in kelvins, and the y -axis represents volume, in liters. Therefore, the estimated volume, in liters, of the hydrogen when its temperature is **500** kelvins is represented by the y -coordinate of the point on the graph that has an x -coordinate of **500**. The point on the graph with an x -coordinate of **500** has a y -coordinate of **7**. Therefore, the estimated volume, in liters, of the hydrogen when its temperature is **500** kelvins is **7**.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.